Gemma 4 Cross-Layer KV-Sharing Topology in Decoupled Cross-Attention

Dedicated BF16 Cross-KV projections exist strictly on KV-Owner layers. KV-Shared layers reuse stationary (K_cross, V_cross) states from designated Anchor layers.

Sliding KV-Owner (Dedicated Cross-KV) · Global Full KV-Owner (512d Head Dim) · Anchor Layer (Cached Cross-KV Provider) · KV-Shared Layer (Reuses Anchor C

PANEL A: GEMMA 4 E2B (35 Layers · 15 KV-Owners / 20 Shared)

Cross-KV Cache Reduction: 57.14% (20 / 35 Layers)

L00	L01	L02	L03	L04	L05	L06	L07	L08	L09	L10	L11	L12	L13	L14	L15	L16	L17
Slide Own	Slide Own	Slide Own	Slide Own	Full Own	Slide Own	Slide Own	Slide Own	Slide Own	Full Own	Slide Own	Slide Own	Slide Own	Slide Anc	Full Anc	Shared	Shared	Shared

L18	L19	L20	L21	L22	L23	L24	L25	L26	L27	L28	L29	L30	L31	L32	L33	L34
Shared	Shared	Shared	Shared	Shared	Shared	Shared	Shared	Shared	Shared	Shared	Shared	Shared	Shared	Shared	Shared	Shared

★ DESIGNATED ANCHORS
L13: Sliding (w=512)
L14: Global Full (512d)

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20 Downstream Layers (L15–L34) completely bypass Cross-KV Projections
Sliding layers reuse L13 Cross-KV; Global layers (L19, L24, L29, L34) reuse L14 Cross-KV. States computed once during prefill.
Result: 57.14% reduction in cross-attention KV-cache memory footprint with zero quality degradation.

PANEL B: GEMMA 4 E4B (42 Layers · 24 KV-Owners / 18 Shared)

Cross-KV Cache Reduction: 42.86% (18 / 42 Layers)

L00	L01	L02	L03	L04	L05	L06	L07	L08	L09	L10	L11	L12	L13	L14	L15	L16	L17	L18	L19	L20
Slide Own	Slide Own	Slide Own	Slide Own	Slide Own	Full Own	Slide Own	Slide Own	Slide Own	Slide Own	Slide Own	Full Own	Slide Own	Slide Own	Slide Own	Slide Own	Slide Own	Full Own	Slide Own	Slide Own	Slide Own

L21	L22	L23	L24	L25	L26	L27	L28	L29	L30	L31	L32	L33	L34	L35	L36	L37	L38	L39	L40	L41
Slide Own	Slide Anc	Full Anc	Shared	Shared	Shared	Shared	Shared	Shared	Shared	Shared	Shared	Shared	Shared	Shared	Shared	Shared	Shared	Shared	Shared	Shared

★ DESIGNATED ANCHORS
L22: Sliding (w=512)
L23: Global Full (512d)

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18 Downstream Layers (L24–L41) completely bypass Cross-KV Projections
Sliding layers reuse L22 Cross-KV; Global layers (L29, L35, L41) reuse L23 Cross-KV. States computed once during prefill.
Result: 42.86% reduction in cross-attention KV-cache memory footprint with zero quality degradation.